Standard Operating Procedure for Resolving Huawei Access Node JAY-ACC-60 ASIC Memory Parity Errors and Line Card Crashes at Jayanagar Metro Site

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2666

Description

The Huawei Access Node JAY-ACC-60 is a critical component of the Jayanagar Metro Site's telecommunications infrastructure, and as such, it is imperative to thoroughly understand the underlying protocol and hardware mechanics that lead to ASIC memory parity errors and line card crashes. The Access Node's Role in the Huawei network architecture is to provide secure, high-speed data transmission services to end-users. The ASIC (Application-Specific Integrated Circuit) is responsible for processing and forwarding data packets across the network. In the event of an ASIC memory parity error, the line card may crash, resulting in a loss of connectivity and service disruptions. This SOP outlines the steps necessary to diagnose and resolve such issues on the Huawei Access Node JAY-ACC-60. Understanding the root causes of these failures is crucial in preventing future occurrences and ensuring the overall reliability and uptime of the network. The ASIC memory parity errors and line card crashes are primarily caused by hardware or environmental faults, which can be attributed to a range of factors including power supply redundancy loss, temperature fluctuations, or manufacturing defects. By following this SOP, network engineers can effectively troubleshoot and resolve these issues, minimizing downtime and ensuring the continued reliability of the network.

Pre-requisite Checks

- Dispatch a team of remote hands to the Jayanagar Metro Site to perform physical checks and verify the site's environmental conditions.

- Verify the status of the redundant link to ensure it is functioning correctly.

- Check the power supply redundancy to confirm it is operating within normal parameters.

Potential Root Causes

Cause	Probability
Power Supply Redundancy Loss	High
Temperature Fluctuations	Medium
Manufacturing Defects	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include ASIC, Expected Output: Current configuration for ASIC memory parity errors and line card crashes, Description: This command displays the current configuration of the ASIC, including any relevant settings or parameters that may be contributing to the memory parity errors and line card crashes.

Command: debug ASIC memory parity error, Expected Output: Debugging information for ASIC memory parity errors, Description: This command enables debugging for ASIC memory parity errors, providing detailed information about the error conditions and helping engineers isolate the root cause of the issue.

Command: show system redundancy, Expected Output: Redundancy status for power supply and line cards, Description: This command displays the current redundancy status for the power supply and line cards, helping engineers identify any potential issues with the redundant link or power supply redundancy.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace ASIC/line card, Description: If the diagnostic commands indicate that the issue is caused by a faulty ASIC or line card, this step involves replacing the affected component with a new one.

Step 2: Step: Reconfigure the power supply redundancy, Command: configure system redundancy power-supply, Description: If the issue is caused by a power supply redundancy loss, this step involves reconfiguring the power supply redundancy to ensure that the system can continue to operate in the event of a power supply failure.

Step 3: Step: Adjust the environmental conditions, Command: configure system environment, Description: If the issue is caused by temperature fluctuations, this step involves adjusting the environmental conditions to ensure that the system operates within a stable temperature range.

Post-Check Verification

```
Command: display current-configuration | include ASIC, Expected Output: Current configuration for ASIC memory parity errors and line card crashes, Description: This command verifies that the configuration changes made during the resolution steps have successfully resolved the issue.

Command: show system redundancy, Expected Output: Redundancy status for power supply and line cards, Description: This command verifies that the power supply redundancy and line card redundancy are functioning correctly after the resolution steps.
```

Roll-back Procedures

- Step: Revert the configuration changes, Command: undo replace ASIC/line card, Description: If the resolution steps do not resolve the issue, this step involves reverting the configuration changes made during the resolution steps to ensure that the system returns to its original state.

- Step: Restore the original power supply redundancy configuration, Command: undo configure system redundancy power-supply, Description: If the issue was caused by a power supply redundancy loss, this step involves restoring the original power supply redundancy configuration to ensure that the system can continue to operate in the event of a power supply failure.